StockEvent Lens — Batch 002 Intelligence Dashboard

Articles 006–010 Analysis, Vector Embedding Sensitivity & Conflict Thresholds

Batch ID
`batch_02 (Articles 006-010)`

TEST 006
Original — 100% Base
Nvidia weighs $10B anchor investment in Anthropic $100B IPO at $2T valuation.

TEST 007
Repeat — 96% Overlap
Re-syndicated fixture wire story with zero new facts or state changes.

TEST 008
Update — +22% New Info
Moves talks into formal due diligence & virtual data room access.

TEST 009
Conflict — Dispute ($5B)
Contradicts initial ceiling: claims maximum cap was $5B, not $10B.

TEST 010
Different — 0% Overlap
Unrelated announcement of Singapore Robotics Lab starting Group 004.

Technical Analysis: Dense Vector Blindness vs. Numerical Contradiction Metrics

A core challenge in automated financial news aggregation is that dense vector models (e.g., BAAI/bge-small-en-v1.5, OpenAI Ada-002) compute cosine similarity based on *semantic context* rather than *numerical fidelity*. Changing financial figures (e.g., $10 billion vs $5 billion ceiling in test_009) yields high vector similarity (~0.89) because both texts share identical surrounding context tokens ("Nvidia", "Anthropic", "anchor investment", "IPO").

```
// Formula: Formal Conflict Detection Condition
Conflict_State = ( Cosine_Similarity(E_orig, E_cand) > τ_sim ) && ( ΔF_contradiction == 1.0 )
where τ_sim = 0.60, and ΔF_contradiction = 1.0 indicates a hard numerical or directional negation.
```

1. Dense Embedding Blindness
Dense embeddings project words into high-dimensional space where "$10 Billion" and "$5 Billion" share near-identical context embeddings. Thus, pure vector cosine distance fails to flag this metric conflict.

2. LLM Evaluation Override
To prevent missing critical valuation disputes, the LLM reasoning layer performs explicit entity-attribute-value matching, overriding high vector similarity when $\Delta F_{contradiction} = 1.0$ is detected.

Article Relationship & Percentage Breakdown (Batch 002)

FILE IDENTIFIER	PATTERN LABEL	KEYWORDS IDENTIFIED	SIMILARITY %	DELTA %	CLASSIFICATION RATIONALE
`test_006_original`	Original (%5=1)	Nvidia, Anthropic, $10B investment, $100B IPO raise, $2T valuation	100%	Baseline	Establishes breaking anchor investment news frame.
`test_007_repeat`	Repeat (%5=2)	Anchor investment, Anthropic IPO, $10B ceiling, $2T valuation	96%	0% New Facts	Rephrased wire report without new factual developments.
`test_008_update`	Update (%5=3)	**Due diligence**, data room access, financial records review	84%	+22% Operational Facts	Advances event state from preliminary talks to active due diligence.
`test_009_conflict`	Conflict (%5=4)	**$5B ceiling dispute**, competing account, original report challenge	89% Match	$\Delta F_{contradiction} = 1.0$	Directly disputes initial $10B ceiling figure, asserting $5B cap.
`test_010_different`	Different (%5=0)	Singapore lab, robotics simulation, university partners, 2027 start	0%	Group 004 Trigger	Zero domain overlap; triggers standalone Group 004 event cluster.

Group 003 — Consolidated Event Summary

4 Articles (006–009)

NVIDIA is reportedly weighing an anchor investment in AI startup Anthropic's potential initial public offering, as the Claude maker considers raising up to $100 billion at a valuation of approximately $2 trillion before the November U.S. midterm elections (Articles 006–007). Discussions have progressed into due diligence, with NVIDIA receiving access to a virtual data room containing financial records (Article 008). However, sources dispute the potential investment ceiling: original reports cited a cap of up to $10 billion, whereas a competing account claims the initial ceiling considered in discussions was $5 billion (Article 009 ; $\Delta F_{contradiction} = 1.0$). Talks remain preliminary, and no definitive investment agreement has been signed.

✓ Update Preserved: Data Room Due Diligence ✓ Conflict Preserved: $10B vs $5B Ceiling
✓ Duplicate Removed: Article 007 Consolidated

Group 004 — Standalone Summary

1 Article (010)

NVIDIA has announced a robotics research laboratory in Singapore established in partnership with local universities. The facility will focus on simulating warehouse robotics prior to physical deployment, with its inaugural research program scheduled to launch in January 2027 (Article 010).

✓ New Event Cluster Established ✓ Zero Entity Contamination from Group 003

Batch 002 Raw Data Source Inspector

JSON Format Verification & Evidence Code Snippets

test_006_original.json — Group 003

```
{
  "id": 61752792,
  "headline": "Nvidia Eyes Up To $10 Billion Investment
  "symbols": ["AMZN", "GOOG", "GOOGL", "NVDA"],
  "summary": "Nvidia may invest up to $10 billion in A
}
```

Establishes $10B cap baseline.

test_008_update.json — Group 003

```
{
  "id": 990000008,
  "headline": "Nvidia And Anthropic Move Proposed IPO
  "symbols": ["AMZN", "GOOG", "GOOGL", "NVDA"],
  "summary": "Nvidia has begun reviewing Anthropic's f
}
```

Adds due diligence & data room access.

test_009_conflict.json — Group 003

```
{
  "id": 990000009,
  "headline": "Report Disputes $10 Billion Ceiling...",
  "symbols": ["AMZN", "GOOG", "GOOGL", "NVDA"],
  "summary": "A competing account says Nvidia's investr
}
```

Disputes $10B vs $5B ceiling.